

FramSat-1 Downlink Reception & Telemetry Decoding Guide

Technical Specifications for Amateur Radio Operators

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Abstract

This document provides the radio frequency (RF), modulation, framing, and telemetry specifications for the UHF downlink of the **FramSat-1** CubeSat, developed by Orbit NTNU. It is intended to assist amateur radio operators, SatNOGS ground station operators, and enthusiasts in receiving, demodulating, and decoding telemetry transmitted by the satellite.

1 Introduction

FramSat-1 is a 1U student satellite developed by Orbit NTNU in Trondheim, Norway. The satellite transmits periodic beacon and telemetry frames on the amateur satellite 70 cm band. All transmissions adhere strictly to open amateur radio conventions, utilizing direct synchronous HDLC-framed packets modulated with standard G3RUH 9600 bps GFSK.

Amateur radio operators are invited to listen for downlinks and submit reception reports to the Orbit NTNU ground operations team.

2 Physical Layer (RF & Modulation)

The physical layer parameters for the UHF downlink are summarized in Table 1.

Table 1: FramSat-1 Downlink RF Specifications

Parameter	Value / Specification
Downlink Frequency	435.141 MHz (IARU Coordinated)
Frequency Band	70 cm Amateur Satellite Service (435–438 MHz)
Modulation Scheme	GFSK (2-FSK with Gaussian filter)
Bitrate / Baudrate	9600 bps
Frequency Deviation (Δf)	± 3.0 kHz
Modulation Index (h)	≈ 0.625
Pulse Shaping Filter	Gaussian ($BT = 0.5$, G3RUH standard)
Scrambling Scheme	Standard G3RUH ($1 + x^{12} + x^{17}$, Mask: 0x21)
Bit Encoding	NRZI (Non-Return-to-Zero Inverted)
Preamble Length	50 bytes, nominal (0x7E / alternating synchronization bytes)
Postamble Length	7 bytes

3 Data Link Layer & Frame Structure

FramSat-1 utilizes synchronous HDLC framing with standard flag delimiters (0x7E) and a 16-bit CRC-CCITT Frame Check Sequence (FCS) implemented by the radio hardware.

Downlink frames are transmitted directly from the onboard computer without AX.25 callsign address encapsulation. The transmission begins with a 6-byte transport/network header, followed immediately by the FramSat-1 housekeeping telemetry payload (beginning with the static ASCII signature "FS1.0"). The over-the-air frame layout is summarized in Table 2.

Table 2: FramSat-1 Over-The-Air Frame Structure

Byte Range	Length	Field	Description
0–5	6 B	Transport Header	Onboard network routing header (0x85 0x00 0x85 0x00 0x00 0xCF)
6–10	5 B	Signature	Static synchronization signature: "FS1.0"
11–24	14 B	Housekeeping Data	System health (ID, mode, RSSI, telecommands, reboots, battery, uptime)
25+	168–180 B	Sensor Payload	Attitude sensor arrays (168 B zero-padding in LEOP mode; up to 180 B nominal)
Trailer	4 B	CRC / Trailer	32-bit frame check sequence

3.1 Frame Synchronization & Length

Total frame length is dynamically managed by the onboard flight software:

- **LEOP Mode (Deployment):** Frame length is exactly 197 bytes (attitude sensors unpowered / zero-padded to 168 bytes).
- **Nominal Mode:** Frame length is up to 209 bytes (9 complete 20-byte sensor measurement samples).

Receivers should search for the 5-byte ASCII sequence "FS1.0" (0x46 0x53 0x31 0x2E 0x30). In nominal downlinks, this signature appears at byte index 6 following deframing.

4 GNU Radio Companion (GRC) Receiver Pipeline

A complete demodulation chain can be built in GNU Radio Companion utilizing the standard `gr-satellites` and `gr-osmosdr` out-of-tree modules (both included by default in the recommended Radioconda distribution). Figure 1 illustrates the recommended flowgraph utilizing a standard RTL-SDR receiver.

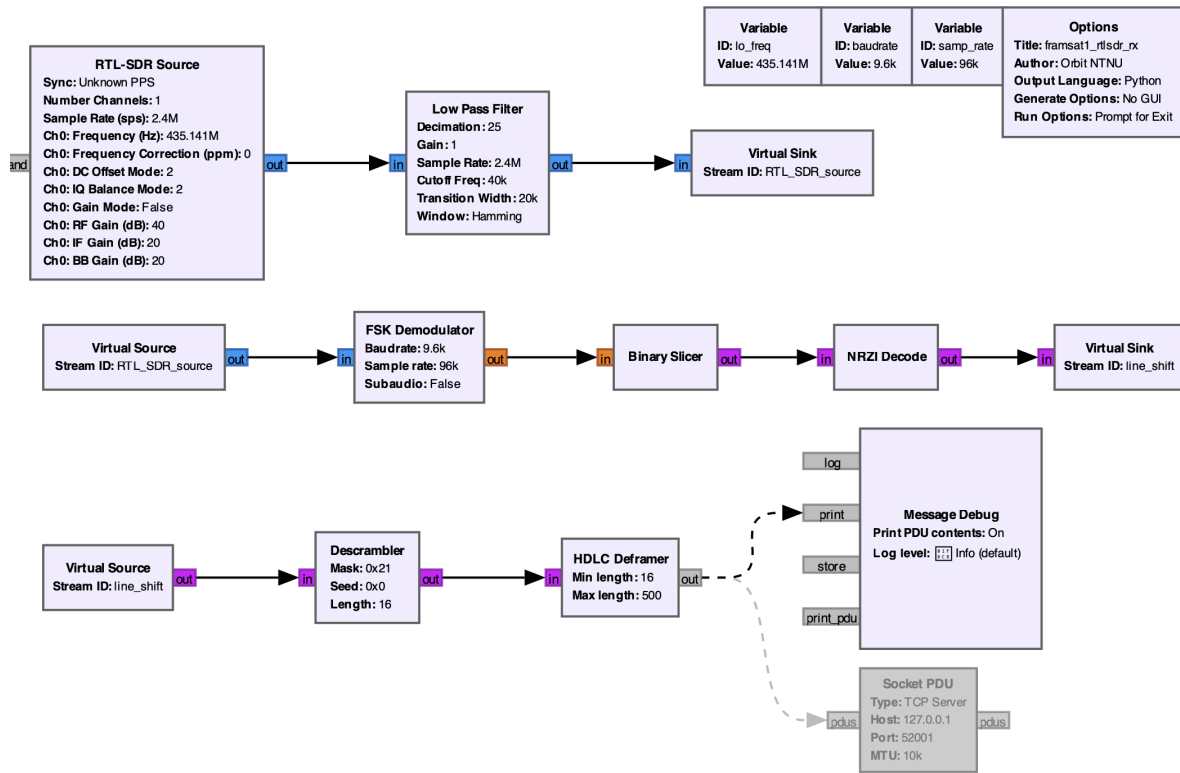


Figure 1: Recommended GNU Radio receiver pipeline for FramSat-1 using an RTL-SDR dongle (sampling at 2.4 MSps and decimated to 96 kSps).

The recommended demodulator architecture consists of:

1. **RTL-SDR Source:** Center frequency set to 435.141 MHz, sample rate 2.4 MSps, and RF Gain set to 40 dB.
2. **Low Pass Filter (Decimator):** Decimates by a factor of 25 (reducing the rate from 2.4 MSps to 96 kSps), with a cutoff frequency of 40 kHz and transition width of 20 kHz to suppress out-of-band noise.
3. **FSK Demodulator:** Operates at the decimated sample rate of 96 kSps with baud rate set to 9600 bps.
4. **Binary Slicer:** Converts demodulated audio-frequency excursions into binary symbols (0 and 1).
5. **NRZI Decode:** Decodes bit transitions according to standard G3RUH / NRZI conventions.
6. **Descrambler:** Mask: 0x21, Seed: 0x0, Length: 16 (Standard G3RUH polynomial).
7. **HDLC Deframer:** Minimum frame length: 16 bytes, Maximum: 500 bytes. Validates the 16-bit CRC-CCITT (FCS) and discards corrupt frames.
8. **Message Debug / Socket PDU:** Outputs valid received frames to the console via **Message Debug**, or optionally streams them to external software via **Socket PDU** (TCP Server on port 52001).

5 Reference Frame Decoder

To assist the amateur radio community, an open-source Python reference decoder is maintained in the project repository:

<https://github.com/orbitntnu/framsat1-telemetry-decoder>

The script (`framsat1.decoder.py`) provides two operational modes:

- **Live Network Mode (`--listen`):** Connects directly to GNU Radio via TCP (port 52001) to ingest, deframe, and display satellite telemetry in real time during an active pass.
- **Offline / Inspection Mode:** Accepts a raw hexadecimal frame string as a command-line argument to parse historical or crowdsourced frames recorded by other stations.

The decoder dynamically identifies the "FS1.0" synchronization signature, parses the frame header, and unpacks the full housekeeping telemetry structure into human-readable engineering units.

6 Housekeeping Telemetry Format

FramSat-1 transmits periodic health and status telemetry within the received frame, beginning with the 5-byte ASCII signature "FS1.0" (located at byte offset 6 following the 6-byte transport header). All multi-byte numeric fields are serialized in standard **Big-Endian** format (network byte order, MSB first). The layout of the telemetry payload is specified in Table 3 (offsets are relative to the "FS1.0" signature):

Table 3: FramSat-1 Housekeeping Telemetry Format

Offset	Size	Field	Type	Description / Scaling
0–4	5 B	signature	ASCII String	Static sync signature: "FS1.0"
5	1 B	beacon_id	uint8_t	Packet sequence counter
6	1 B	operating_mode	uint8_t	Mode: 1 = Default, 2 = LEOP
7	1 B	satellite_rssi	uint8_t	Ground uplink RSSI recorded by satellite
8–9	2 B	telecommand_count	uint16_t	Total telecommands accepted by satellite
10	1 B	eps_flags	uint8_t	EPS status and subsystem power flags
11–12	2 B	reboot_count	uint16_t	Total OBC / EPS reboot counter
13–14	2 B	battery_voltage	uint16_t	Battery voltage ($V = \text{val}/1000.0$ V)
15–18	4 B	uptime_sec	uint32_t	Satellite uptime in seconds
19+	168–180 B	sensor_data	Array	Sensor data (168 B LEOP; 180 B nominal)

6.1 Identification Signature

Ground stations can reliably identify FramSat-1 downlinks by verifying that the received payload contains the ASCII signature "FS1.0" (at byte offset 6 of the deframed data).

7 Submitting Reception Reports

Orbit NTNU greatly appreciates reports and raw telemetry recordings from the radio community. If you receive packets from FramSat-1, please submit:

- Your callsign and ground station location (grid square or coordinates).
- UTC timestamp of reception.
- Raw hexadecimal packet string or IQ baseband recording (`.wav` / `.sigmf`).
- Estimated SNR or waterfall screenshot.

Reports can be submitted via email to satcom@orbitntnu.com or uploaded to the SatNOGS network database under satellite **FramSat-1**.